DEPD Executive Analytics & GIS Dashboard

Final Development Specification

Version 1.3 (Final) | Supersedes Version 1.2 | Ready for Antigravity / Streamlit development

Government of Sindh — Department of Empowerment of Persons with Disabilities (DEPD)

Source schema	Business definitions	Status
Kobo XLSForm shared 4 Nov 2026 (id_string: depd_beneficiary_data_v6; 296 survey rows, 215 choices). The 7 Nov export has 1,510 columns, so the live form is newer: re-share it (decision 26) and a 107-row real sample export (1 to 7 Nov 2026). Live source: Kobo export-settings API (Section 9)	DEPD Beneficiary Data Collection Form, Version 7 (Word)	Final for development. Beneficiary rules: 5.3 to 5.5. Summary Reports: 5.7. Map: Section 7. Live data and deployment: Sections 9 and 12.

1. Executive objective

Build a secure, responsive, visually polished, live Streamlit executive dashboard for aggregate beneficiary and service reporting by DEPD facilities and partner organisations. The latest Kobo XLSForm is the exact technical schema, and the sample export shows how users really fill it. The Version 7 Word form is the business-definition reference. This document specifies the dashboard only; it does not change or redeploy the Kobo form.

Out of scope: individual beneficiary records, cross-facility person-level deduplication, financial or grant spending analysis, and any change to the Kobo form.

2. Source audit (verified against the latest XLSForm and the sample export)

Finding	Observed result	Implementation consequence
Workbook structure	survey: 296 rows (header excluded), 33 groups, 143 integer fields, 29 multi-select questions, 1 calculation. choices: 215 rows. No repeat groups.	One submission = one flat row = one facility for one reporting period.
Form identity	settings: id_string = depd_beneficiary_data_v6, version = 7.0.	Use the deployed asset UID, not the document title. Store form version with every row.
Geography	6 divisions and 30 districts. Fields division and district; district is filtered by division in the form.	Join district to division from the choice list. Flag any mismatch (Section 11).
Facility identity	facility_name is free text. facility_type has 19 choices (Section 6), including the new gia, sindh_persons_with_disabilitie, dis_cer and legal_support__beneficiaries.	Create a governed facility registry with stable IDs and alias matching. Never rely on names alone.
Geolocation	The form has one geopoint question named GIS in Section A. It is now required (41 of 43 sample reports have a value; Kobo also exports _GIS_latitude and _GIS_longitude). It is officer-reported and unverified.	Use registry coordinates first. Use reported GIS as "Reported, not yet verified" only after the checks in Section 7.
Headline headcounts	w_male, w_female, w_trans, w_other, w_total (calculation), w_new, w_waiting. Section W is hidden for pcrdp, assistive_ip and regional_dir.	pcrdp and regional_dir are excluded from beneficiary KPIs ("Not applicable"); assistive_ip uses its device recipients (5.3, decision 23).

Finding	Observed result	Implementation consequence
Section W reliability (sample)	In 43 real reports: 3 have W = 0 while their section holds real active figures; 2 copied the cumulative total into W (W 607 vs c_active 78; W 422 vs 65); 7 differ from the section figure by more than 20%; 3 are all zero.	Never sum w_total alone. Resolve Direct beneficiaries with the ladder in Section 5.3.
Indirect reach fields	New: Total_indirect_reach_since_inception (Section C), Beneficiary_students_inception_In_Direct (Deaf Reach), Total_Indirect_Reach_ablishment_Indirect (legal support).	Reported indirect reach is separate from the family estimate (Section 5.5).
New facility types and groups	gia (Grant-in-Aid), sindh_persons_with_disabilitie (SPDPA), dis_cer (Issuing District, disability certificates), legal_support__beneficiaries (Access to Justice). New groups: disability certificate (G1C) and legal support (G2). New question Who_is_the_primary_r_d_materials_services (Outreach / Internal).	Added to routing (Section 6) and the metric dictionary (Appendix A).
Sample data quality	12 "DEPD Digital Classes Deaf Reach" reports are filed as gov_se; facility names are free text with spelling variants; contact_name and officer_designation sometimes hold a phone number or 0; no report has a Kobo validation status yet.	Diagnostics in Section 11.3. Never auto-correct stored values.
New / waiting coverage	w_new and w_waiting are asked only for deaf_reach, sipmr, carts, hostel, vtc, academy, ipmr and nowpdp. rcmhc, rcph and gov_se use c_new and c_waiting. inclusive uses n_new and n_waiting.	Use the canonical coalesce rules in Section 5. A KPI built on w_new alone would miss five facility types.
Age and disability coverage	w_age_groups (6 choices) and w_disability are multi-select.	Coverage only. Never chart them as population distributions.
Reporting period	rp_start is now labelled "Since Date of Center/Institution Inception" and rp_end is the report date. 8 of 43 sample reports have rp_start equal to rp_end. There is no quarter or month field.	Use rp_end only for the period (year, quarter, month). Treat rp_start as the facility inception date. Cadence is unconfirmed (Section 14).
Verification	Two different things: form field verification_status (verified / not_verified, set by the reporting side) and Kobo _validation_status (set by reviewers in Kobo).	Treat a row as accepted only when Kobo _validation_status is Approved, if the review workflow is used. Show both statuses separately.
Assistive devices	Section R is shown to all facility types, including NOWPDP. NOWPDP also reports devices in P5.	Real double-count risk. One canonical source per submission (Section 5.2).
Personal / sensitive fields	contact_name, contact_info, signature (image), s_pic_1 to s_pic_5 (images), s_story_file (PDF).	Access-controlled. Never shown on public or viewer pages (Section 11).
Document discrepancy	The V7 Word document header still says "Version 5".	Confirm the canonical version name before publication.
Regional Director	regional_dir holds office information only (Sections A and B). The office manages several services and has no beneficiaries of its own.	Not a service. Excluded from service lists, totals, breakdowns and service counts. Shown only in an admin Offices view and as a green dashed office marker (7.1).
Ownership vs type	In the 107-row export, 37 of 58 gov_se reports say ownership = partner (mostly the "DEPD Digital Classes Deaf Reach" centres); Deaf Reach is mostly "other"; C-ARTS is government in all 5.	Ownership is self-declared and unreliable. Colour by the DEPD class of the service type; the submitted value is a check (7.1).

Finding	Observed result	Implementation consequence
Implausible values	One RCMHC report has a cumulative value of 3.37e215.	Sanity ceiling per field (default 1,000,000, admin setting): exclude from sums and flag "implausible value".
Data availability	The XLSForm holds schema only. Submissions arrive through the Kobo export-settings API (Section 9).	The app reads live data through a cache. Synthetic fixtures are used only for tests and are labelled.

3. Dashboard architecture and layout

Visual style: modern executive government analytics. Navy #18345A, teal #008C95, white #FFFFFF, slate #66758C. Wide desktop layout, responsive tablet and mobile layout, left sidebar navigation and a compact top filter bar. DEPD and Sindh government logos only if approved assets are supplied.

Page	Required features
1. Executive Overview	Reach summary block (Direct, Estimated family, Estimated total; Active / Cumulative toggle; Section 5.3 to 5.5), then reporting facilities, coverage %, new, waiting, reported indirect reach and devices distributed. Also time trends, gender donut, service mix, division comparison, mini coverage map and latest-submission status.
2. Geographic Coverage	Neutral division polygons, district boundaries, district choropleth, facility pins coloured by ownership (green government, blue partner organisation, purple other) with service icons (Section 7), the headline and service panels and the Data received panel (7.4).
3. Beneficiaries & Gender	Reach summary; unique active, new and waiting; male / female / transgender / other; trends; breakdown tables by facility type, region (division) and district (Section 5.6). Age and disability coverage only.
4. Service Performance	Education, rehabilitation and health, training, employment, assistive devices, identity facilitation, residential. Correct units for each indicator (Appendix A).
5. Partner Programs	NOWPDP P1 to P5, C-ARTS V1 to V5, Deaf Reach (T), SIPMR (U), Inclusive Education (N). Drill-down to programme KPIs.
6. Facility Directory & Profile	Searchable facilities with status, district, services, reporting history, completeness, aggregate metrics and map point. Detail tabs in Section 8.
7. Pivot & Reports	Group by division / district / facility / service / period, metric selector, valid aggregation per the metric dictionary, CSV / XLSX, PNG and PDF export.
8. Summary Reports	Which divisions, districts and service types have updated data; the Summary Sheet listing every service with region, district, organisation name and beneficiaries; district x service matrix; pending list; Excel and JPG downloads (Section 5.7).
9. Data Quality & Diagnostics	Coverage denominators, late and missing reports, missing coordinates, region-district mismatch, duplicates, anomalies, API sync status.

4. Global cascading filters (mandatory)

- **Hierarchy:** Division → District → Facility. Start at All Sindh / All Districts / All Facilities. A division limits districts and facilities. A district limits facilities. Child selections are cleared when the parent changes.
- **Independent filters:** reporting year, quarter / month or date range; service type (multi-select); ownership class (Government / Partner organisation / Other); facility type; service category; programme (NOWPDP P1 to P5, C-ARTS V1 to V5); ownership; operational status (op_status); Kobo validation status; form verification status.

- **The same filter bar is on every tab, including the map:** service type, ownership class (government / partner organisation / other), region (division), district and facility, plus period and the Active / Cumulative toggle. Changing a filter on one tab keeps it on the others until Reset All.
- **One filtered dataset.** Every filter updates KPIs, map layers, charts, pivots, exports and facility cards from the same canonical filtered dataframe. Show active-filter chips and a Reset All button.
- **Multi-select** for comparisons, with an optional single-selection drill-down mode.
- **Selected-period logic:** snapshot indicators use the latest valid submission per facility in the selection. Flow indicators sum over distinct reporting periods (by rp_end). Cumulative indicators use the latest snapshot per facility and are never summed across periods. Use the Asia/Karachi time zone for all period boundaries.
- **Labelling:** totals across facilities are labelled "sum of facility-reported unique headcounts". Person-level deduplication across facilities is not possible from aggregate data.

5. KPI definitions and reporting semantics

5.1 Canonical KPIs

KPI	Source fields	Aggregation rule
Reporting facilities	Registry facility_id with an accepted submission	Distinct count for the selected period. Without a registry, use matched names and label the result "unverified".
Expected facilities	External approved facility registry	If unavailable, show "coverage denominator unavailable".
Reporting coverage %	Reporting ÷ expected eligible facilities × 100	Show only when a denominator exists.
Direct beneficiaries (Active)	Resolved per Section 5.3 (w_total first, then the section figure)	Latest valid submission per facility; sum across facilities. Never add snapshots across time. Not a plain sum of w_total.
Indirect (family) estimate	Direct x F, F = 5	Model estimate, always labelled "Estimated". Section 5.4.
Total reach	Direct + Indirect = Direct x (1 + F)	Always labelled "Estimated". Section 5.4.
Reported indirect reach	Total_indirect_reach_since_inception, Beneficiary_students_inception_In_Direct, Total_Indirect_Reach_ablishment_Indirect	Cumulative; separate card; never added to the three numbers above (Section 5.5).
New beneficiaries	coalesce(w_new, c_new, n_new)	Sum over distinct reporting periods (by rp_end).
Waiting list	coalesce(w_waiting, c_waiting, n_waiting)	Latest snapshot per facility. Label "facility-reported, possible cross-facility overlap".
Cumulative served	coalesce(c_cumulative, n_integrated, t_cum, u_cum, p_cum, Total_Direct_Benefic_since_establishment, Cumulative_Disabilit_sued_Since_Inception)	Latest snapshot per facility. Not collected for hostel, vtc, academy, ipmr or carts (V1 to V5 overlap): show "Not collected". Used by the Cumulative basis (Section 5.3).
Gender	w_male, w_female, w_trans, w_other	Latest snapshot per facility, used only when W passed the check in 5.3 (otherwise "Gender not reported"), then sum over the filtered scope.
Devices distributed	Section 5.2 mapping	Distributed quantities only. Never add Section U manufacturing counts.

KPI	Source fields	Aggregation rule
Service outputs	Appendix A, unit column	Keep people, sessions, visits, devices, items, km/day, schools, cases and audits separate. Never total across units.

5.2 Device mapping (one canonical source per submission)

Canonical device	Section R field(s)	NOWPDP P5 field(s)
Wheelchairs	r_wheelchair	p5_wheelchair
Crutches, sticks, walkers	r_crutches + r_walker	p5_crutches
Motorised tri-wheelers	(not collected)	p5_tricycle
Prosthetic limbs	r_limb	(not collected)
Orthoses	r_orthosis	(not collected)
Hearing aids	r_hearing	p5_hearing
Vision aids (spectacles, low-vision aids, white canes)	r_spectacles	p5_cane
Other devices	r_dev_other	p5_other

- **Rule:** for a nowpdp submission use P5 quantities. For every other facility type use Section R quantities when r_given = yes. Never add the two.
- **Flag** "device double-count risk" when a nowpdp submission has r_given = yes and any R quantity above zero. Use P5 only until a reviewer resolves it.
- **Recipients** (r_male, r_female, r_trans, r_other and p5_cum / p5_fitted) are people. Devices are quantities. Never mix them. Section U fields (u_prosth_made, u_orth_made, u_repaired) are workshop output and stay in a separate measure.

5.3 Direct beneficiaries: resolving each submission

Many facilities leave Section W empty, type 0, or copy the cumulative total into it, and enter the real numbers in their own section. The dashboard therefore resolves one Direct figure per submission, with a recorded source and confidence. Reference code: tests/reference_resolution.py.

Inputs: W = w_total; S = the section-level active figure for the facility type (table below); C = the cumulative figure.

Step	Condition	Result
1	Facility type is pcrdp or regional_dir (office only, not a service)	Not applicable (not 0). assistive_ip has no W: its Direct is the people who received devices in the period (R male + female + transgender + other), flagged "period count, not stock".
2	W above 0 and W is not a copy of C (W equals C while S exists and S differs)	Use W (source "W"). If S above 0 and W differs from S by more than 20%, keep W and flag "review" (show both).
3	Otherwise S above 0	Use S (source "section"). Flag W_blank, W_zero or W_copied_from_cumulative.
4	Otherwise W = 0 or S = 0	Direct = 0, flag all_zero_report (low confidence). Never silently dropped.
5	Otherwise	No data.

facility_type	S (section active figure)	C (cumulative figure)
`rcmhc, rcph, gov_se, gia, pcrdp`	c_active	c_cumulative
`inclusive`	n_enrolled	n_integrated
`carts`	Largest of v1_active to v4_active (lower bound, low confidence). V5 parents are excluded.	none
`nowpdp`	Largest of p2_active, p3_active (lower bound, low confidence)	p_cum
`deaf_reach`	none	t_cum
`sipmr`	none	u_cum
`dis_cer, sindh_persons_with_disabilitie`	none	Cumulative_Disabilit_sued_Since_Inception
`legal_support__beneficiaries`	none	Total_Direct_Benefic_since_establishment
`hostel, vtc, academy, ipmr`	none	none (show "Not collected")

- **Basis toggle:** Active (default) uses the steps above. Cumulative uses C only (first non-blank); W is never used there. Dis_cer, SPDPA and legal-support facilities appear on the Cumulative basis.
- **Aggregation:** latest valid submission per facility and period (facility_id + rp_end); Direct per facility, then summed across facilities. Label totals "sum of facility-reported unique headcounts".
- **Transparency:** under each card show the source mix (for example "34 facilities from Section W, 5 from section data, 3 zero reports, 7 need review"). Clicking opens the facility list with W, S, the resolved value, the source and the flags.
- **Worked check:** on the 43 sample reports (before de-duplication) a plain sum of w_total gives 2,882, but the resolved Direct total is 2,205 (5 reports resolved from section data, 3 all-zero, 7 flagged for review). Family estimate 11,025; total reach 13,230. Expected values are in tests/expected_sample_summary.json.

5.4 Indirect (family) estimate and total reach

Number	Formula	Label
Direct beneficiaries	Resolved unique persons (5.3)	Direct (facility-reported)
Indirect (family) beneficiaries	Direct x F, F = 5	Estimated, Direct x 5
Total reach	Direct + Indirect = Direct x (1 + F)	Estimated total reach

- F is an admin-only setting (default 5). Show F in the card tooltip and in every export.
- The family estimate is a model, not a count. The word "Estimated" must appear wherever it or the total is shown. Never present it as reported data.
- Not part of Direct, and therefore not multiplied: C-ARTS V5 parent trainees, and any device, session, visit, case or certificate count.

5.5 Reported indirect reach (separate)

Facilities also report indirect reach themselves (outreach and community): Total_indirect_reach_since_inception, Beneficiary_students_inception_In_Direct and Total_Indirect_Reach_ablishment_Indirect. This is a different measure from the family estimate. Show it as its own cumulative card. Never add it to Direct, to the family estimate or to Total reach, because the two can describe the same people. The Outreach / Internal answer to

Who_is_the_primary_r_d_materials_services is reported separately; Internal recipients are already inside the facility's own count and are never added to Direct.

5.6 Breakdown by facility type, region and district

Executives need the same numbers cut three ways. Provide three breakdown tables, each reconciling to the grand total, with the Active / Cumulative toggle and the Reach summary (5.3 to 5.5).

View	Grouped by	Rules
By facility type (service)	facility_type as filed; optional "with suggested corrections" view	Default is "as filed". A banner shows the number of suggested corrections (for example 12 Deaf Reach reports filed as gov_se). The corrected view uses only reviewed overrides, never name matching alone. List all 18 service types (regional_dir is office information only), including those with no reports ("No reports yet").
By region (division)	Division derived from the district using the 30-district list	Never use the submitted division field for totals (flag mismatches). Always list all 6 divisions; a division with no reports shows "No reports", not 0.
By district	district	Always list all 30 districts with their division. Click a district to set the global filter. Districts with no reports are greyed "No report".

Columns in every breakdown table: reports, facilities (distinct), Direct, Indirect (family, Estimated), Total reach (Estimated), new, waiting, cumulative, male / female / other (only from reports where W was usable), and a source-mix flag (how many rows used W, the section figure, zero reports or are flagged for review).

- **Cross views:** service x division heatmap of Direct, and division -> district -> facility drill-down.
- **Reconciliation:** the rows of each table must add up to the grand total of the Reach summary under the same filters. Add a test for this.
- **Sort and share:** sortable columns and a share-of-total column; blank and zero stay different.
- **Reference results:** tests/expected_sample_107.json gives the expected numbers for the current 107-row export (regional_dir excluded: 106 service reports, 27 of 30 districts, all 6 divisions): Direct 7,728; Indirect (x5) 38,640; Total 46,368. By division: Sukkur 2,086; Hyderabad 2,066; Karachi 1,991; Shaheed Benazirabad 864; Larkana 607; Mirpurkhas 114. tests/expected_breakdown.csv is the older 43-row sample.

5.7 Summary Reports tab: who has updated data

One tab answers "which region, district and service types have updated data, and how many beneficiaries". It follows the global filters, the reporting period (rp_end, not submission time) and the Active / Cumulative toggle. Regional Director offices are excluded.

Block	Content
Coverage strip	Divisions updated x of 6; districts updated x of 30; service types updated x of 18; facilities reporting; reports received; last submission (Asia/Karachi).
Division table	All 6 divisions: status (Updated / No report), districts and services updated, reports, Direct, Indirect (Estimated), Total reach (Estimated), last update.
District table	All 30 districts with division, status, service types that reported (icon chips), reports, Direct, Indirect, Total, last update. "No report" rows stay visible in grey.
Service-type table	All 18 services: status, divisions and districts covered, reports, Direct, Indirect, Total, new, waiting.
Matrix	District (rows) x service type (columns); cell = Direct; "-" = no report; toggle to report counts.
Pending list	Districts and services with no report in the period; CSV export for follow-up.

Block	Content
Summary Sheet (register)	One row per service report: Service type, Service category, Ownership class, Region (division), District, Organisation / institute name, Reporting period, Direct, Indirect (Estimated), Total reach (Estimated), Source flag. Sorted by service type, then region, then district, with a subtotal per service type and a grand total. The organisation name is facility_name (for assistive_ip the partner name in r_org). Institution names only: no contact or personal fields.

- Every table reconciles to the grand total and has Download Excel and Download JPG buttons (Section 8). Reference: tests/expected_summary_report.csv and tests/expected_summary_sheet.csv (106 rows; government 76 reports, 4,803 Direct; partner 30 reports, 2,925 Direct).

6. Facility types, routing and service categories

The form has 19 facility types; 18 are services and regional_dir is office information only. Derive service-category flags from the union of the facility_type default, services_available, carts_services and nowpdp_programs. Use the facility-type default alone when the others are empty.

facility_type	Service category (map icon)	Sections shown in the form
rcmhc	Rehabilitation and education (multi-service)	C, R, W
rcph	Rehabilitation (medical cross)	C, R, W
gov_se	Education / school (school building); subtype recorded in M	C, M, R, W
inclusive	Education (book)	N, R, W
deaf_reach	Education / school (school building)	T, R, W
sipmr	Health and rehabilitation; assistive devices	L, U, R, W
ipmr	Health and rehabilitation (medical cross)	L, R, W
carts	Training and therapy, multi-service (tools)	V1 to V5 as selected, Q if partner support, R, W
nowpdp	Multi-service (layered icon); identity, education, vocational, employment, devices	P0 to P5 as selected, W (R discouraged, see 5.2)
hostel	Hostel / residential (bed)	I, R, W
vtc	Vocational (tools)	K, R, W
academy	Teacher training (graduation cap)	H, R, W
gia	Grant-in-Aid, inclusive education (book with hand)	C, H, I, K, L, M, N, R, W
sindh_persons_with_disabilitie	Registration, complaints and certificates (identity card)	G1 (+ certificate group if ticked), R, W (W shown: decision 14)
dis_cer	Disability certificates (identity card)	Certificate group, R, W (W shown: decision 14)
legal_support__beneficiaries	Legal support, Access to Justice (identity card)	Legal support group, R, W (W shown: decision 14)
pcrdp	Registration (identity card). Legacy type.	C, R. No Section W and no G1 (G1 now belongs to SPDPA: decision 16).
assistive_ip	Assistive devices (wheelchair)	R only. No Section W.
regional_dir	Office information only (not a service). Green dashed office marker.	A and B only. No beneficiary counts. Excluded from all service views.

7. Sindh GIS specification

Use authoritative, licensed Sindh district boundary GeoJSON or shapefiles (30 districts) and derived division boundaries. Validate administrative names against the XLSForm choices. Never fabricate polygons.

7.1 Marker colour (ownership class), shape (service category) and icon (service)

Colour = ownership class, fixed by service type (rule from DEPD, data/ownership_by_type.csv). The submitted ownership field is only a check.

Class	Colour	Hex	Service types
Government	Green	#2E7D32	RCMHC, RCPH, government special education, C-ARTS, vocational training centre, SPDPA, disability certificate (dis_cer), inclusive education, Regional Director
Partner organisation	Blue	#1F5FBF	Assistive devices, GIA, Deaf Reach
Other	Purple	#7B3FA0	Remaining types: SIPMR, IPMR, hostel, academy, NOWPDP, PCRDP, legal support (proposed; to confirm)
Unknown	Grey	#6B7280	Facility type missing or not in the table

Shape = service category, so health, education and training differ even without colour:

Category	Shape	Icon inside
Health / rehabilitation (RCMHC, RCPH, SIPMR, IPMR)	Circle	Medical cross
Education / school (gov_se, inclusive, Deaf Reach, GIA)	Rounded square	School building (gov_se, Deaf Reach), book (inclusive), book with hand (GIA)
Training / vocational (C-ARTS, VTC, Academy, NOWPDP)	Hexagon	Tools (C-ARTS, VTC), graduation cap (Academy), briefcase (NOWPDP)
Hostel / residential	House (pentagon)	Bed
Assistive devices	Triangle	Wheelchair
Registration / legal (SPDPA, disability certificate, legal support, PCRDP)	Diamond	Identity card, scales
Regional Director (office, not a service)	Square, dashed outline	Office building

Shape, icon, colour and a text label are always shown together, and the legend lists all four. A multi-service facility keeps its primary shape and adds a small badge. Division polygons use neutral grey tints (#EEF1F4 and #E3E8ED) with dark outlines, so green, blue and purple belong to ownership alone. Check icon contrast (at least 4.5:1) and test the colours with a colour-vision simulator.

Submitted ownership is a check. When the submitted ownership differs from the class of the type (for example 37 of 58 gov_se reports say partner; 20 of 25 Deaf Reach say other), keep the class colour and show "submitted ownership differs" in Data Quality. A reviewed facility-level override wins when supplied. In the "with suggested corrections" view, a gov_se report named Deaf Reach moves to deaf_reach and turns blue. Never recolour silently.

7.2 Layers and behaviour

- **Layers:** divisions, districts, service coverage, beneficiary choropleth, facility markers (qualifying coordinates only), marker clustering and an optional density heatmap.

- **Service icons and shapes:** see 7.1 and docs/gis_icon_map.csv. Regional Director offices appear as a green dashed office marker in the admin Offices view and are never counted as a service. Icons describe facility services, not people.
- **District hover:** division, district, reporting facilities, active / new / waiting counts, service availability and coverage %. Clicking a district applies the global district filter.
- **Facility click:** official name, type, location, status, reporting period, unique active / new / waiting, gender breakdown, services and a link to the facility profile. No individual records.
- **Export:** JPG of the complete Sindh extent every time (not the current zoom), with the filtered markers, title, filters, period, legend, source, last-refreshed timestamp and disclaimer; HTML optional (Section 8).

7.3 Facility coordinate rules

Source	Status label	Rule
Registry approved coordinates	Verified	Preferred. Always used when present.
Submission GIS geopoint (also _geolocation)	Reported, not yet verified	Use only if the registry has none, the point lies inside Sindh and inside the polygon of the submitted district, and the latest submissions agree. Otherwise flag "Missing / invalid location".
No usable coordinates	Missing location	Show the district polygon shading and a labelled facility list. Never place pins at guessed locations or district centroids.

7.4 Panels beside the map

Panel	Content
Headline	Main total beneficiaries (Direct); Total indirect beneficiaries (Estimated, family x 5); Total reach (Estimated).
By service type	One card or bar per service (all 18; "No reports" where empty): Direct, reports, male / female / other where W was usable, new, waiting.
Data received	Reports received, distinct facilities, districts reporting x of 30, last submission time (Asia/Karachi), source mix (W / section / zero / review), and the latest 10 reports (facility, type, district, period, Direct). No contact or personal fields.
Filters	Service type, ownership class, region and district filters sit above the map and apply to markers and panels.
Selection	Clicking a marker or a service card filters every other panel.

All panels follow the global filters and the Active / Cumulative toggle.

8. Charting, drill-downs and exports

- **Charts:** Plotly line / area for trends, grouped bars for district and division comparison, donut for gender, horizontal bars for service outputs, district × service heatmap, and conditionally formatted tables. Provide an accessible table alternative for every chart and map.
- **Trends:** use compatible period definitions. Show missing periods as missing, not zero. Show period-over-period change only for comparable facility cohorts, or label the cohort change.
- **Facility detail tabs:** Overview, Beneficiaries, Services, Program-specific Metrics, Reporting History, Data Quality, Map. Tooltips show definitions and units.
- **Pivot builder:** rows division / district / facility / service; columns period, facility type or programme; values from the metric dictionary. Aggregation follows the rule in Appendix A (latest snapshot, sum of periods, or count distinct).

- **Every table can be downloaded as Excel (.xlsx) and as an image (.jpg).** Each table (breakdowns, Summary Reports, Summary Sheet, pivots, data quality lists, facility directory) has a Download Excel and a Download JPG button. The files contain exactly the rows and columns on screen under the active filters, with a title, the active filters, reporting period, Active / Cumulative basis, source and last-refreshed time (Asia/Karachi). The JPG shows the first 60 rows and says so; the Excel file holds all rows. Reference code: src/export/exporters.py.
- **Charts** download as JPG / PNG at 2x or 3x resolution (Plotly / Kaleido). A management PDF (title, date, filter chips, legend, definitions) is also offered. Exports respect the viewer role and never include restricted fields.
- **Map download:** every map export is a JPG of the complete Sindh extent (total bounds of the approved boundary file plus a margin), regardless of the current zoom or pan. It shows the filtered markers (service type, ownership class, region, district), the legend (colour, shape, icon), title, filters, period, source and refreshed time. The image is rendered on the server (GeoPandas / Matplotlib) so it does not depend on the browser view.

9. Data ingestion and transformation

- **Live source (primary):** the saved Kobo export setting, an XLSX export of all submissions: https://kf.kobotoolbox.org/api/v2/assets/agqkhZsLaqUQyLK8UwKrsx/export-settings/esqPMKgPTD8ZYEERvWqhw9K/data.xlsx . Read it with HTTP GET and the header Authorization: Token <token> unless the export is set to anonymous access in Kobo. The URL and token are configuration (environment variables or Streamlit secrets: KOBO_EXPORT_URL, KOBO_API_TOKEN), never hard-coded, so the source can change without code changes and the app can move to another server.
- **Export settings must not change silently:** the saved setting fixes the column names (XML names, no group prefixes, select_multiple expanded as field/choice 0/1 columns, all versions off). The loader checks required columns on every load and stops with a clear schema error if the export settings were changed.
- **Load behaviour:** the export is generated on request and can take several seconds. Use a timeout (default 120 s), up to 3 retries with back-off, a cache (default 15 minutes) and a visible "Last refreshed" stamp. If the call fails, keep serving the last good copy, marked "stale", and fall back to a local XLSX in data/ only for tests. Never log the token or the full URL with credentials.
- **Strip restricted columns at load:** the export contains contact, signature, picture and story fields. Remove them from the general dataframe immediately and keep them only for admin and manager views.
- **Optional alternative:** the v2 JSON data endpoint (/api/v2/assets/{uid}/data/?format=json) with pagination, if the XLSX export becomes too slow. Same canonical mapping.
- **Source adapter:** map exported columns to canonical names using Appendix A (the current export has no group prefixes; tolerate them, for example sec_w/w_total, if the export settings change), and keep raw values for audit. Read cells as text first and convert numbers explicitly, so blank stays distinct from 0.
- **Kobo system fields to keep:** _id, _uuid, _submission_time, _validation_status, _geolocation, start, end, and the form version. Use start / end for a completion-time quality check.
- **Geography:** join district to division using the 30-district list. Flag mismatches. Analytics use the validated mapping. Keep the original for diagnostics.
- **Facility registry** (governed, separate): stable facility_id, normalised name, facility_type, district, division, approved location, expected reporting status, optional services, and reviewed alias mappings.
- **Deduplication key:** facility_id + rp_end (rp_start is the facility inception date, not a period start). The latest approved submission wins when a facility registry gives a reliable facility_id. Without one, do NOT deduplicate by name: in the 107-row export, 12 groups (32 reports) share one centre name, district and month (several Deaf Reach satellite schools and digital classes), and removing them would lower Direct from 7,728 to 7,112. Count every submission, flag each group for review, and let an admin mark reviewed duplicates. Keep all revisions in an audit view. Recommended form change: add a centre name or code (decision 24).

- **Multi-select fields:** normalise space-separated codes or expanded 0/1 columns into coverage flags. Treat blank (not applicable) differently from 0.
- **Derived tables:** fact_facility_period, dim_facility, dim_district, dim_division, dim_service, fact_service_output, submission_quality. Use pandas and NumPy. Avoid a heavy database at first.
- **Testability:** adapters run on local XLSX / CSV / JSON fixtures. Production credentials live in secrets, not in an editable API-entry form in the app.

10. Programme-specific drill-downs

Programme	Required drill-down	Key fields
NOWPDP	P1 Shanakht: applications, CNICs issued, pending. P2 Taleem: enrolment and packages. P3 Hunar: active, completed, dropout, courses. P4 Rozgar: jobs and self-employment. P5 Suvaari: recipients and device quantities.	p_cum; p1_*; p2_*; p3_*; p4_*; p5_*; nowpdp_programs
C-ARTS	V1 education, V2 therapy sessions, V3 vocational, V4 adult / ADHD livelihoods, V5 parent training. Parent trainees are excluded from the disability beneficiary total.	carts_services; v1_* to v5_*; q_*
SIPMR	OPD, unique therapy patients versus sessions, prosthetics and orthotics made / fitted / repaired, waiting.	l_*; u_*
Deaf Reach	Cumulative and active students, jobs, transport riders, km/day, teacher training, PSL vocabulary.	t_*
Inclusive Education	Integrated and enrolled students, schools, graduates, master-teacher training, monitoring visits.	n_*
Other facilities	RCMHC / RCPH (C), government special education (C, M), PCRDP (G1), hostel (I), academy (H), vocational (K), assistive device partners (R).	c_*; m_*; g1_*; i_*; h_*; k_*; r_*

11. Privacy, security, accessibility and quality

11.1 Privacy and security

- **Aggregate data only.** No names, CNICs, personal phone numbers, individual disability history or beneficiary-level markers.
- **Restricted fields:** contact_name, contact_info, signature, pictures and story files are visible only to admin and manager roles. Pictures appear only where a data-protection approval exists.
- **Roles:** admin, manager, regional coordinator, viewer. Enforce region and facility permissions on the server (data filtered before it reaches the page). Hiding widgets is not security.
- **Hosting:** confirm that the Streamlit Community Cloud, or an alternative, meets DEPD data-residency and access policy. Use real sign-in (for example OIDC) rather than a shared password. See Section 14.

11.2 Accessibility

- WCAG 2.1 AA goals, keyboard navigation, descriptive chart labels, readable contrast, icons plus text, no colour-only encoding, and table alternatives for every map and chart.

11.3 Diagnostics (Data Quality page)

Check	Rule
Missing or invalid geography	Missing district; district not in the 30-district list; division does not match district.
Unrecognised facility	No registry match or alias for facility_name.

Check	Rule
Duplicate periods	Same facility with the same rp_end, or two reports in the same month or quarter.
Stale or late reports	No submission for the expected period; submission_date after the agreed deadline.
Impossible values	Negative numbers; rp_end before rp_start; submission_date before rp_end; rp_end in the future.
Active above capacity	w_total above capacity, or v1_active above v1_cap.
Active versus cumulative	Active above the cumulative figure in the same programme (c_active > c_cumulative, n_enrolled > n_integrated, p2_active > p2_integrated, and the V1 to V5 pairs).
W zero or blank with section data	w_total is 0 or blank while the section figure (c_active, n_enrolled, v-section active) is above 0. Direct resolved from the section (5.3).
W copied from cumulative	w_total equals c_cumulative while c_active differs. Direct resolved from the section.
W versus section mismatch	Difference above 20%. W kept; flagged "review" with both numbers.
All-zero report	Direct = 0 from W and the section. Flag; if devices were given in the same report, mark "probably missing data".
Misclassified facility type	Name contains "deaf reach" but type is gov_se (12 in the sample). Show suggested type; apply only through a reviewed override.
Near-duplicate facility names	Same district, type and rp_end with similar normalised names (case, punctuation, "&" versus "and", typos).
New above active	New beneficiaries above the active figure (for example new 500 with active 152; new 90 with active 1). Possible but unusual: flag for review.
Division mismatch	Submitted division differs from the division of the district. Totals use the district mapping.
Contact field misuse	contact_name, contact_info or officer_designation holds only digits or "0". Restricted fields; flag only.
Period meaning	rp_start equal to rp_end, or rp_start later than the facility's first report. rp_start is treated as the inception date.
Active headcount mismatch	w_total differs from the section-level active figure (c_active, n_enrolled, v-section active) for the same facility.
Gender total	w_total is not the sum of the four gender fields (calculation tampered with or form edited).
Device double-count risk	nowpdp submission with P5 and R quantities both above zero.
C-ARTS parents	V5 parent counts included in a disability beneficiary total.
Location	No coordinates; GIS point outside Sindh or outside the submitted district.
Verification	Form verification_status = not_verified, or Kobo _validation_status not Approved.
Completion time	Very short start-to-end duration (for example under 60 seconds) on a long submission.
API / sync	Failed or partial refresh, schema change, unknown new field.

No fictitious KPI figures. Show "No data", "Not applicable", "Not collected", "Not yet verified" and "Missing location" as distinct states.

12. Development stack and implementation plan

Python 3.10+; Streamlit, pandas, NumPy, Plotly, GeoPandas / Shapely, Folium or PyDeck (streamlit-folium where needed), requests / httpx, openpyxl, XlsxWriter, Matplotlib, Pillow, Kaleido and reportlab. Simple modular architecture with caching. Suggested cache time 15 minutes, with a manual refresh for admins only.

Phase	Acceptance requirement
1. Data contract	Read the XLSForm. Produce a mapping CSV / JSON for all fields (starting from Appendix A). Verify a Kobo API export against the schema, including group prefixes and select_multiple format.
2. Core data layer	Facility registry, 30 districts / 6 divisions, latest-period selection, duplicate handling, canonical KPIs (Section 5), unit tests. Test: coalesce rules return the right source for all 18 services, and the Direct / family / total rules reproduce tests/expected_sample_107.json and tests/expected_kpis.json.
3. Executive dashboard	Responsive cards, charts, synchronised division / district / facility filters, period selector. Test: every number on screen reconciles with the filtered dataframe.
4. GIS	Correct polygons and colours, category icons, coordinate rules (7.3), hover and click drill-down, export.
5. Programme dashboards	NOWPDP, C-ARTS, SIPMR, Deaf Reach, Inclusive Education and other facility metrics; service pivots with correct aggregation.
6. Security and QA	Roles, restricted fields, Data Quality page (11.3), exports, mobile tests, accessibility check, deployment instructions.

12.1 Build, run and deployment (portable)

- **Build:** Antigravity builds the code from this specification and the prompts. The project folder is the source of truth for rules and tests.
- **Run:** Python 3.11 and Streamlit. The app reads only configuration from environment variables or .streamlit/secrets.toml, so the same code runs on a laptop, an office server or a cloud host.
- **Portable deployment:** the folder includes a Dockerfile, docker-compose.yml, .env.example and docs/deployment.md. Moving to another server means copying the folder, setting the environment variables (Kobo URL and token, sign-in settings) and running docker compose up -d.
- **Sign-in and roles:** viewer, regional coordinator, manager and admin, enforced on the server. Put the app behind HTTPS (reverse proxy such as Nginx or Caddy) and the department's sign-in method (decision 8).
- **Operations:** health check, structured logs without secrets, cache refresh by admin only, and a nightly snapshot of the last good dataset so the dashboard still opens if Kobo is unreachable.

13. Developer prompt

The full, paste-ready prompt is prompts/dashboard_developer_prompt_v2.md in the project folder. Summary:

Act as a senior Python / Streamlit engineer, GIS analyst and public-sector data architect. Implement the DEPD Executive Analytics & GIS Dashboard described in this document (Version 1.3).

- Use the latest Kobo XLSForm as the exact schema and the Version 7 Word form for business definitions. Do not invent data, fields, coordinates, beneficiary identities or boundaries. Compare the form with Appendix A and report any difference before coding.
- **Do not trust w_total alone.** Resolve Direct beneficiaries with the rules in 5.3 (W first, then the section figure when W is blank, zero or copied from cumulative), implemented as tested library code that matches tests/reference_resolution.py.
- Show the Reach summary: Direct, Indirect (family) = Direct x 5, and Total reach = Direct + Indirect, on an Active / Cumulative toggle. F = 5 is an admin setting. The estimate is always labelled "Estimated". Reported indirect reach is a separate card and is never added.

- Read live data from the Kobo export-settings URL (Section 9) using KOBO_EXPORT_URL and KOBO_API_TOKEN from configuration. Treat regional_dir as office information only (not a service). Colour map markers by the DEPD ownership class of the service type (green, blue, purple), use a different shape for health, education and training, and add service icons (Section 7.1).
- Use rp_end for periods; treat rp_start as the inception date. Never auto-correct stored values: use reviewed override tables.
- Implement the canonical KPI rules (Section 5), device mapping (5.2), facility routing (Section 6), coordinate rules (7.3) and diagnostics (11.3) exactly as written.
- Deliver modular production-quality code, requirements.txt, .streamlit/config.toml, an example secrets template, unit tests (52 cases in tests/test_cases.md), a setup README and sample data clearly labelled synthetic.
- If the Kobo token, facility registry or boundary GeoJSON is missing, use a clearly marked placeholder and explain the missing dependency. Never invent it.

14. Decisions needed before production

#	Decision	Why it matters	Default if no answer
1	Kobo server, API token permissions, deployed asset UID, and whether fields carry group prefixes.	Needed to connect.	Placeholder; fixture data.
2	Approved district / division boundary dataset and facility registry with verified coordinates.	Map accuracy and coverage %.	District-only map; no pins.
3	Reporting cadence (monthly, quarterly, other), approval workflow and the expected-facility denominator.	Trends and coverage %.	Derive from rp_end; no coverage %.
4	Canonical version name (Word header says "Version 5").	Document consistency.	Use "Version 7".
5	Is a true age distribution required? That needs numeric age counts in the Kobo form.	The form has age-group checkboxes only.	Coverage view only.
6	Confirm the device source rule in 5.2, or tell NOWPDP to leave Section R empty.	Avoids double counting.	P5 for nowpdp, R otherwise.
7	Is the sum of facility-reported headcounts acceptable, or is an external deduplication source needed?	Credibility of totals.	Label as facility-reported.
8	Hosting and sign-in method that meets DEPD data policy.	Security and residency.	Private deployment with sign-in.
9	English only, or English and Urdu labels in the dashboard.	Usability for staff.	English only.
10	Total reach: Direct + Direct x 5, or Direct x 5 only?	Changes the headline number.	Direct + Direct x 5.
11	Family factor F.	Changes the estimate.	5, admin setting.
12	Should a facility's own reported indirect reach replace the x5 estimate for that facility?	Possible overlap of the same people.	No, shown separately.
13	When W and the section figure differ by more than 20%, which is right?	Changes Direct for 8 of 107 sample reports.	Keep W, flag for review.
14	Should dis_cer, SPDPA and legal-support facilities fill Section W?	They only have cumulative figures; W is not meaningful for them.	Yes for now; Cumulative basis.
15	Is rp_start the facility inception date? Should the form get a separate period-start question?	Needed to detect overlapping reports.	Inception date; period from rp_end.

#	Decision	Why it matters	Default if no answer
16	PCRDP: still shows Section C but no G1. Is SPDPA replacing PCRDP?	Wrong questions for PCRDP users.	Treat as built; flag.
17	Rename auto-generated field names (group_dp8mz64, Cumulative_Disabilit_..., legal_support__beneficiaries) before collecting more data?	Readable exports; renaming later breaks columns.	Use as exported.
18	Service breakdown default: "as filed" or "with suggested corrections"?	Deaf Reach reports filed as gov_se change service (20 in the 107-row export).	As filed, with a banner and a toggle.
19	Ownership class by service type (government: RCMHC, RCPH, government special education, C-ARTS, VTC, SPDPA, disability certificate, inclusive, Regional Director; partner: assistive devices, GIA, Deaf Reach) and confirmation of the purple types.	Submitted ownership is unreliable.	DEPD rule in data/ownership_by_type.csv; submitted value is a check.
20	Kobo export access: token owner, anonymous access or not, and who may change the export setting.	A changed export setting breaks columns.	Token in secrets; schema check on every load.
21	Hosting target and sign-in for the live deployment (Docker on a DEPD or cloud server).	Security and portability.	Docker, private network, sign-in.
22	Regional Director is office information only.	Avoids counting an office as a service.	Excluded from services and totals.
23	Do Assistive Devices (assistive_ip) recipients count as Direct?	Changes Direct by 208 in the 107-row export.	Yes, flagged as a period count.
24	Same centre name in the same district and month: duplicate or different centres? Add a facility ID or centre code to the form?	32 reports in 12 groups; Direct 7,728 vs 7,112 if all were duplicates.	Count all; flag groups; admin marks reviewed duplicates.
25	SPDPA row filed by an NGO (Ida Rieu): wrong facility type?	Direct 1,054 sits under a registration type.	Keep value, flag "type may be wrong".
26	The form has grown: about 660 export columns are not in the field dictionary (including a rewritten assistive-device section R1 to R5, still empty). Share the latest XLSForm.	Appendix A may miss new fields.	Loader ignores unknown columns with a warning; Phase 1 reports differences.

Appendix A. Metric dictionary (generated from the XLSForm of 4 Nov 2026)

The live form has since grown (about 660 export columns are not listed here, including the rewritten assistive-device section R1 to R5, still empty in all reports). Phase 1 must regenerate this appendix from the newest XLSForm and report differences. Unknown columns never break the load.

Every number field in the form (143, plus the calculated w_total). Unit "people (indirect reach)" fields are never added to Direct. Some Total_* legal-support fields have no period hint in the form: they are assumed cumulative (CUM) except Total_Ongoing_Cases (SNAP); confirm and add hints. **Rule codes:** CUM = latest cumulative snapshot per facility; SNAP = latest snapshot at reporting end date; FLOW = sum over distinct reporting periods. Section letters match the form. Regenerate this table if the form changes.

Field	Indicator	Unit	Rule
Section B — Operational profile			
`capacity`	Approved beneficiary capacity (if applicable)	seats / places	FLOW
Section C — RCMHC, RCPH, Govt SE, GIA			
`c_cumulative`	Total unique beneficiaries served since establishment	people	CUM
`Total_indirect_reach_since_inception`	Total indirect reach since inception	people (indirect reach)	CUM
`c_active`	Active beneficiaries currently enrolled / receiving treatment	people	SNAP
`c_waiting`	Unique beneficiaries on active waiting list	people	SNAP
`c_new`	New beneficiaries registered	people	FLOW
`c_completed`	Beneficiaries completing rehabilitation / education	people	FLOW
`c_rcmhc_multi`	RCMHC: Children receiving multidisciplinary rehabilitation	people	FLOW
`c_rcph_mobility`	RCPH: Patients receiving mobility training	people	FLOW
`c_rcph_fitted`	RCPH: Unique beneficiaries fitted with prosthetic / orthotic devices	people	FLOW
Section G1 — SPDPA registration & complaints			
`g1_received`	Registrations received	registrations	FLOW
`g1_approved`	Registrations approved	registrations	FLOW
`g1_rejected`	Registrations rejected	registrations	FLOW
`g1_pending`	Registrations still pending	registrations	SNAP
`g1_issued`	Certificates / ID cards issued	certificates / cards	FLOW
`g1_complaints`	Complaints received	complaints	FLOW
`g1_resolved`	Complaints resolved	complaints	FLOW
Section G1 — Disability Certificate group			
`Cumulative_Disabilit_sued_Since_Inception`	Cumulative Disability Certificates Issued Since Inception	certificates	CUM
`Cumulative_Disabilit_Female_Beneficiaries`	Cumulative Disability Certificates Issued to Female Beneficiaries	certificates	CUM

Field	Indicator	Unit	Rule
Cumulative_Disabilit_o_Male_Beneficiaries	Cumulative Disability Certificates Issued to Male Beneficiaries	certificates	CUM
Cumulative_Disabilit_gender_Beneficiaries	Cumulative Disability Certificates Issued to Khwajasira / Transgender Beneficiaries	certificates	CUM
Number_of_certificat_o_children_under_18	Number of certificates issued to children (under 18)?	certificates	CUM
Number_of_certificat_issued_to_adults_18	Number of certificates issued to adults (18+)?	certificates	CUM
Section H — Training Academy			
h_completed	Trainees completed / certified	people	FLOW
h_dropped	Trainees dropped out	people	FLOW
h_trainers	Trainers present	people	SNAP
Section K — Vocational			
k_completed	Trainees completed	people	FLOW
k_dropped	Trainees dropped out	people	FLOW
k_placed	Trainees placed in jobs / self-employment	people	FLOW
Section L — Medical / Physical Rehabilitation			
l_opd	OPD visits	visits	FLOW
l_assess	Assessments done	assessments	FLOW
l_referrals	Referrals made	referrals	FLOW
Section N — Inclusive Education			
n_integrated	Beneficiaries integrated into mainstream schools	people	CUM
n_enrolled	Beneficiaries enrolled in inclusive classrooms	people	SNAP
n_waiting	Beneficiaries on inclusive education waiting list	people	SNAP
n_schools	Mainstream inclusive schools registered / identified	schools	SNAP
n_graduated	Students with disabilities graduated / passed out	people	FLOW
n_tot	Mainstream master teachers trained in inclusive education (ToT)	people	FLOW
n_new	New students with disabilities integrated into mainstream schools	people	FLOW
n_visits	Inclusive education monitoring visits conducted	visits	FLOW
Section Q — Partner Report (C-ARTS)			
q_male	People reached — Male	people	FLOW
q_female	People reached — Female	people	FLOW
q_trans	People reached — Transgender	people	FLOW
q_other	People reached — Other / not recorded	people	FLOW
Section R — Assistive Devices			

Field	Indicator	Unit	Rule
r_male	People who received devices — Male	people	FLOW
r_female	People who received devices — Female	people	FLOW
r_trans	People who received devices — Transgender	people	FLOW
r_other	People who received devices — Other / not recorded	people	FLOW
r_wheelchair	Wheelchair (manual / electric)	devices	FLOW
r_crutches	Crutches / walking sticks	devices	FLOW
r_walker	Walker / rollator	devices	FLOW
r_limb	Artificial limb (prosthesis)	devices	FLOW
r_orthosis	Orthosis / calipers / braces	devices	FLOW
r_hearing	Hearing aid	devices	FLOW
r_spectacles	Spectacles / low-vision aid	devices	FLOW
r_dev_other	Other devices	devices	FLOW
Section T — Deaf Reach			
t_cum	Beneficiary students enrolled since program inception	people	CUM
Beneficiary_students_inception_In_Direct	Beneficiary students enrolled since program inception In-Direct	people (indirect reach)	CUM
t_placed	Graduates placed in permanent jobs	people	CUM
t_await_job	Graduates currently awaiting employment placement	people	SNAP
t_transport	Active students using pick-and-drop transport	people	SNAP
t_km	Total daily distance covered by all transport vehicles (KM/day)	km per day	SNAP
t_teachers	Educators trained as specialized deaf teachers	people	FLOW
t_psl	Digital vocabulary words documented in PSL Dictionary	words	CUM
Section U — SIPMR			
u_cum	Unique patients rehabilitated since establishment	people	CUM
u_discharged	Patients safely discharged after completing rehabilitation	people	FLOW
u_physio_pts	Unique physiotherapy patients served	people	FLOW
u_physio_ses	Physiotherapy sessions conducted	sessions	FLOW
u_ot_pts	Unique occupational therapy patients served	people	FLOW
u_ot_ses	Occupational therapy sessions conducted	sessions	FLOW
u_speech_pts	Unique speech therapy patients served	people	FLOW
u_speech_ses	Speech therapy sessions conducted	sessions	FLOW
u_prosth_made	Custom prosthetic devices manufactured	devices	FLOW
u_orth_made	Custom orthotic devices manufactured	devices	FLOW
u_fitted	Unique beneficiaries fitted with prosthetic / orthotic devices	people	FLOW
u_repaired	Prosthetic / orthotic devices repaired or adjusted	devices	FLOW

Field	Indicator	Unit	Rule
`u_await_dev`	Patients awaiting prosthetic / orthotic devices	people	SNAP
Section V1 — C-ARTS Special Education			
`v1_cum`	Student beneficiaries enrolled since establishment	people	CUM
`v1_active`	Student beneficiaries currently enrolled	people	SNAP
`v1_wait`	Child beneficiaries on active education waiting list	people	SNAP
`v1_cap`	Approved enrolment capacity (classroom seats)	seats / places	SNAP
`v1_exit`	Beneficiaries completing or exiting the program	people	FLOW
Section V2 — C-ARTS Therapy			
`v2_cum`	Therapy beneficiaries served since establishment	people	CUM
`v2_active`	Therapy beneficiaries actively receiving treatment	people	SNAP
`v2_wait`	Therapy beneficiaries on active waiting list	people	SNAP
`v2_sessions`	Therapy sessions conducted	sessions	FLOW
Section V3 — C-ARTS Vocational			
`v3_cum`	Vocational beneficiaries trained since establishment	people	CUM
`v3_active`	Vocational beneficiaries currently enrolled	people	SNAP
`v3_wait`	Vocational beneficiaries on active waiting list	people	SNAP
`v3_completed`	Trainees successfully completing training	people	FLOW
`v3_placed`	Trainees placed in jobs or self-employment	people	FLOW
Section V4 — C-ARTS Adult & ADHD			
`v4_cum`	Adult beneficiaries engaged since establishment	people	CUM
`v4_active`	Adult beneficiaries currently participating	people	SNAP
`v4_wait`	Adult beneficiaries on active waiting list	people	SNAP
`v4_livelihood`	Adults in livelihood / employment tracks (café, organic farming, etc.)	people	SNAP
Section V5 — C-ARTS Parent Training			
`v5_cum`	Parent beneficiaries trained since establishment	people	CUM
`v5_active`	Parents currently enrolled in active training cycles	people	SNAP
`v5_wait`	Parents on active waiting list for upcoming cohorts	people	SNAP
`v5_sessions`	Parent training sessions held	sessions	FLOW
Section P0 — NOWPDP summary			
`p_cum`	Total unique beneficiaries served across all programs	people	CUM
Section P1 — Shanakht			
`p1_cnic`	Unique individuals facilitated with Special CNICs	people	CUM
`p1_apps`	New legal registration applications initiated	cases	FLOW
`p1_issued`	Special CNIC cases processed and issued	cases	FLOW
`p1_pending`	Pending verification cases with NADRA / Social Welfare	cases	SNAP
`p1_guidance`	Beneficiaries receiving legal / procedural guidance only	people	FLOW

Field	Indicator	Unit	Rule
Section P2 — Taleem			
p2_integrated	Children with disabilities integrated into classrooms	people	CUM
p2_active	Active students currently enrolled in education programs	people	SNAP
p2_waiting	Out-of-school children with disabilities on active education waiting list	people	SNAP
p2_packages	School Essential Packages (bags, uniforms, books) distributed	items	FLOW
p2_graduated	Students with disabilities passing annual exams / graduating	people	FLOW
Section P3 — Hunar			
p3_cert	Unique trainees certified	people	CUM
p3_active	Active trainees currently attending vocational courses	people	SNAP
p3_waiting	Eligible applicants on active waiting list for training	people	SNAP
p3_completed	Trainees completing trade courses	people	FLOW
p3_dropped	Trainees dropping out / discontinuing training	people	FLOW
Section P4 — Rozgar			
p4_cum	Graduates placed in permanent corporate / banking jobs	people (placements)	CUM
p4_new	Corporate job placements secured	people (placements)	FLOW
p4_await	Trained job-seekers currently awaiting employment placement	people	SNAP
p4_selfemp	Beneficiaries establishing small enterprises / self-employment	people	FLOW
p4_audits	Corporate accessibility audits and sensitization sessions conducted	sessions	FLOW
Section P5 — Suvaari			
p5_cum	Unique individuals provided assistive devices	people	CUM
p5_fitted	Unique beneficiaries fitted with customized mobility devices	people	FLOW
p5_await	People awaiting device measurement / fabrication	people	SNAP
p5_wheelchair	Manual wheelchairs distributed	devices	FLOW
p5_tricycle	Customized motorized tri-wheelers distributed	devices	FLOW
p5_crutches	Crutches / walking sticks / walkers distributed	devices	FLOW
p5_hearing	Hearing aids distributed	devices	FLOW
p5_cane	White canes / low-vision aids distributed	devices	FLOW
p5_other	Other customized devices distributed	devices	FLOW
Section G2 — Legal support (Access to Justice)			
Total_Direct_Benefic_since_establishment	Total Direct Beneficiaries Served unique counts since establishment	people	CUM
Total_Indirect_Reach_ablishment_Indirect	Total Indirect Reach (Beneficiaries) unique counts since establishment (Indirect)	people (indirect reach)	CUM
Total_Ongoing_Cases	Total Ongoing Cases	cases	SNAP

Field	Indicator	Unit	Rule
`Total_Closed_Resolved_Cases`	Total Closed / Resolved Cases	cases	CUM
`Total_Court_Representations_Provided`	Total Court Representations Provided	representations	CUM
`Total_Legal_Advice_Consultations`	Total Legal Advice/Consultations	consultations	CUM
`Total_Mobile_Clinic_Participants`	Total Mobile Clinic Participants	people	CUM
Section W — Global unique beneficiary summary (headline source; see 5.3)			
`w_male`	Male	people	SNAP
`w_female`	Female	people	SNAP
`w_trans`	Transgender	people	SNAP
`w_other`	Other / Not recorded	people	SNAP
`w_total`		people	SNAP
`w_new`	Unique new beneficiaries registered	people	FLOW
`w_waiting`	Unique people on active waiting list	people	SNAP

Note: gender fields in Q (q_male to q_other) and R (r_male to r_other) count people reached or given devices in the period (FLOW). They are not part of the Section W headline headcount.

Appendix B. Changes in Version 1.3 (from 1.2)

Area	Change
Live data	Kobo export-settings XLSX URL as the primary source (Section 9), token and URL from configuration, schema check, cache, stale-copy fallback, restricted columns stripped at load.
Regional Director	regional_dir is office information only: excluded from services, totals and counts; 18 services remain. Admin Offices view and green dashed office marker.
Summary Reports	New Section 5.7 and page 8: who has updated data by division, district and service type, with beneficiaries, matrix and pending list.
Map	Disability certificate is government. Marker colour by ownership class fixed by service type (green government, blue partner organisation, purple other, grey unknown); marker shape by category (circle health, rounded square education, hexagon training, and others); icons; neutral division polygons; submitted ownership kept as a check; panels beside the map (7.4).
Evidence	107-row export: Direct 7,728; Indirect (x5) 38,640; Total 46,368; plain sum of w_total 8,267. assistive_ip counts device recipients. One implausible cumulative value (3.37e215).
New fields	Provincial, Office_Address_Details, and inclusive-education reported indirect categories (parents / family, peer students, community, school staff).
Deployment	Section 12.1: portable Docker deployment, health check, nightly snapshot.
Summary Sheet and exports	Summary Sheet (service type, region, district, organisation name, beneficiaries); Excel and JPG download for every table; map JPG always of the full Sindh extent; service type, ownership class, region and district filters on every tab including the map; disability certificate moved to government.
Decisions	Added 19 to 26.

Appendix C. Changes in Version 1.2 (from 1.1)

Area	Change
Beneficiary counting	New Sections 5.3 to 5.5: Direct is resolved per submission (W first, then the section figure when W is blank, zero or copied from cumulative); Indirect (family) estimate = Direct x 5; Total reach = Direct + Indirect; reported indirect reach kept separate. Active / Cumulative toggle.
Evidence	Source audit now includes a 43-row real sample: 3 W-zero, 2 W-copied-from-cumulative, 7 mismatches, 3 all-zero, 12 misclassified Deaf Reach rows. A plain sum of w_total (2,882) differs from the resolved Direct total (2,205).
Form changes recorded	19 facility types (gia, SPDPA, dis_cer, legal support added); certificate and legal-support groups; indirect reach fields; primary-recipient question; GIS now required; rp_start relabelled as inception date; q_sections removed.
Period logic	Period comes from rp_end only; rp_start is the inception date; overlap detection by rp_start is dropped.
Diagnostics	Added: W zero or blank, W copied from cumulative, W versus section mismatch, all-zero report, misclassified type, near-duplicate names, contact field misuse, period meaning.
Breakdowns	New Section 5.6: tables by facility type, region (division) and district, with reconciliation to the grand total and reference numbers in tests/expected_breakdown.csv.
Decisions	Added decisions 10 to 18 with defaults. Appendix A regenerated (143 number fields).
Project folder	Added the developer prompt v2, reference code, anonymised sample cases with expected results, and 38 test cases.

Appendix D. Changes in Version 1.1 (from 1.0)

Area	Change
Source audit	Corrected: the form does have an optional geopoint (GIS). Added form fields that were missing from the audit (contact, ownership, R, Q, G1, H, I, K, L, M, P). Clarified row counts.
KPIs	Replaced single-field rules with canonical coalesce rules, because w_new and w_waiting are not asked for every facility type. Added cumulative KPI.
Devices	Added an exact R / P5 field mapping and the one-source rule.
Verification	Separated Kobo _validation_status from the form field verification_status.
GIS	Added coordinate source rules and status labels; proposed colour hex values and a colour-blind test.
Routing	Added the 15 facility types with service categories and sections.
Diagnostics	Replaced the generic list with concrete, testable rules using real field names.
Security	Restricted personal fields; hosting and sign-in decision added.
New	Appendix A metric dictionary with unit and aggregation rule for all 127 integer metrics plus w_total; decision list with defaults.